

momentum conservation - practice problems

name _____

1. On a low-friction track, a 500 g cart moves right at 0.80 m/s while a 250 g cart waits at rest. What is the total momentum of the two-cart system?
2. A 4.0 kg cart moves right at 2.0 m/s, and a 2.0 kg cart moves left at 3.0 m/s. What is the total momentum? Let right be positive.
3. A 2.0 kg cart moving right at 3.0 m/s collides with a 1.0 kg cart at rest. After the collision, the 2.0 kg cart moves right at 1.0 m/s. What is the final velocity of the 1.0 kg cart? The collision lasts 0.020 s. What average force does each cart feel?
4. At an amusement park, a 180 kg bumper car (including its rider) moving right at 4.0 m/s hits a 120 kg bumper car moving left at 1.0 m/s. After the collision, the 180 kg car moves right at 2.0 m/s. What is the velocity of the 120 kg car?

5. A 90 kg linebacker running at 5.0 m/s tackles a 75 kg running back who is running straight toward him at 6.0 m/s. They hold on to each other. What is their velocity right after the tackle?
6. A 1.5 kg lump of clay is thrown right at 8.0 m/s. It hits and sticks to a 2.5 kg cart that is rolling left at 2.0 m/s. What is their final velocity?
7. A 30 000 kg railroad car rolling at 2.0 m/s bumps into and couples with a 20 000 kg car rolling in the same direction at 0.50 m/s. How fast do the coupled cars move?
8. A 0.010 kg bullet is fired into a 2.0 kg wooden block resting on ice. The bullet stays in the block, and the block slides away at 2.5 m/s. How fast was the bullet moving?
9. Two carts on a track are held together at rest with a compressed spring between them. When the spring is released, the 2.0 kg cart moves right at 3.0 m/s. What is the velocity of the 3.0 kg cart?

10. A 60 kg hockey player gliding at 5.0 m/s grabs onto a teammate who is standing still on the ice. They glide off together. What is their velocity?

11. A 3000 kg spacecraft is coasting at 4000 m/s. Explosive bolts separate it into a 2000 kg capsule and a 1000 kg empty rocket stage. After the separation, the capsule moves forward at 4500 m/s. What is the velocity of the rocket stage?

12. A 55 kg astronaut is floating at rest 20 m from the space station. To get back, she throws a 2.0 kg wrench directly away from the station at 8.0 m/s. How long does it take her to reach the station?

13. A 70 kg skater standing still on smooth ice catches a 5.0 kg backpack thrown at 6.0 m/s. What is the skater's velocity after the catch?

14. Three air-track gliders collide. A 2.0 kg glider moves right at 3.0 m/s, a 1.0 kg glider moves left at 4.0 m/s, and a 5.0 kg glider is at rest. After the collision, the 1.0 kg glider moves right at 1.0 m/s and the 5.0 kg glider moves right at 0.50 m/s. What is the final velocity of the 2.0 kg glider?

15. A 0.40 kg air-track glider moving right at 0.50 m/s runs into a 0.20 kg glider that is moving right more slowly. After the collision, the 0.40 kg glider moves right at 0.30 m/s and the 0.20 kg glider moves right at 0.60 m/s. How fast was the 0.20 kg glider moving before the collision?

16. On a pool table, a 0.17 kg cue ball moving east at 2.0 m/s hits a 0.17 kg ball at rest. After the collision, the cue ball moves 1.0 m/s east and 0.87 m/s north. What are the east and north velocity components of the other ball?

17. A 200 g toy car moving right at 1.2 m/s sticks to a 300 g toy truck moving left at 0.30 m/s. What is their final velocity?

18. In a simplified crash model, a 900 kg compact car moving at 36 km/h rear-ends a 1500 kg van moving in the same direction at 18 km/h. The two vehicles lock together. Ignore braking during the short collision. What is their speed right after the collision?

19. When you jump, your momentum changes from zero to a large upward momentum. Does that break the law of conservation of momentum?